FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS | GRADE 10 | SUB-STRAND 2.1: SIMILARITY AND ENLARGEMENT	
Student Assessment Document	
Student Name	______________________________________
Class	______________________________________
Date	______________________________________

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation document. Use it as a model to write your own complete, well-reasoned answer to the driving question. Read each section carefully, study the exemplar response, then write your own version using evidence from your investigations, experiments, and classwork across the 12 lessons. Your answer should show that you truly understand the mathematics — not just that you can repeat definitions. Write in full sentences and paragraphs. Use diagrams, calculations, and real examples where they strengthen your explanation.

Section 1 — Answer the Driving Question Directly

Start your explanation by answering the driving question head-on. Why does resizing a passport photo change its size but never its shape? Name the mathematical concept that makes this possible and state it clearly in your own words.	Resizing a passport photo changes its size but never its shape because the photographer's software applies a single multiplying number — called the Linear Scale Factor (LSF) — to every length in the original image equally. When every width, every height, and every diagonal is multiplied by exactly the same number, all the proportions in the image stay the same: the ratio of your forehead height to your face width stays identical, every angle in the image stays the same, and the overall outline of the photo keeps the same rectangular shape. The mathematics that makes this possible is the concept of similarity. Two figures are mathematically similar when one is an exact enlargement or reduction of the other — meaning all corresponding angles are equal and all corresponding lengths are in the same ratio. For the Kimathi Street passport studio, the standard print (35 mm × 45 mm) and the noticeboard print (105 mm × 135 mm) are similar rectangles with a linear scale factor of 3, because 105 ÷ 35 = 3 and 135 ÷ 45 = 3. That single number, 3, scaled every length in the image uniformly — and that is why the face looks perfectly natural at both sizes.

Section 2 — Explain the Conditions for Similarity

Explain what must be true for two figures to be mathematically similar. What are the two conditions? Why must BOTH conditions be satisfied? Use the passport	For two figures to be mathematically similar, two conditions must both be true at the same time: (1) all corresponding angles must be equal, and (2) all corresponding sides must be in the same ratio — that is, they must all share the same linear scale factor. You need both conditions because satisfying only one is not enough. Consider a square and a rectangle: all their angles are 90°, so condition 1 is met — but their sides are not in the same ratio, so they are NOT similar. Now consider the passport photo distortion: when the photographer accidentally resizes only the width of the 35 mm × 45 mm photo — say, doubling it to 70 mm — but leaves the height at 45 mm, the width scale factor is 70 ÷ 35 = 2 while the height scale factor is 45 ÷ 45 = 1. These scale factors are

photo and at least one other Kenyan example to illustrate your answer. Explain also what goes wrong when only one condition is met — why does the face look distorted when only the width is changed?	different, so the sides are no longer in the same ratio. The face is stretched sideways: your cheeks look too wide, your forehead looks too low, and the image becomes unrecognisable. This is a direct, visible consequence of breaking the similarity condition. Another Kenyan example: a maize seed and the fully grown maize cob it produces are NOT mathematically similar, because the cob grows much more in length than in width — the scale factors in different directions are not equal. However, a map of Kenya's Rift Valley drawn at a scale of 1 : 50 000 and the actual landscape it represents ARE similar, because every distance on the map is scaled by exactly the same factor of 50 000 in every direction.

Section 3 — The Linear Scale Factor and the Area Scale Factor	
Explain how the linear scale factor (LSF) and the area scale factor (ASF) are related. Derive or show why the area scale factor is always the square of the linear scale factor. Use the passport photo measurements to calculate both scale factors and verify the relationship with real numbers. Explain what this means practically — for example, how much more paper does the noticeboard print use than the passport print?	The linear scale factor (LSF) is the number you multiply every length by to go from the original figure to the enlarged (or reduced) figure. For the passport studio on Kimathi Street: LSF = 105 ÷ 35 = 3 (you can verify: 135 ÷ 45 = 3 also). The area scale factor (ASF) is the number by which the area is multiplied. Here is why the ASF is always equal to the LSF squared: Area is calculated by multiplying two lengths together (for example, Area = length × width). When you scale a figure by LSF, every length is multiplied by LSF — so the new area is (LSF × original length) × (LSF × original width) = LSF^2 × (original length × original width) = LSF^2 × original area. Therefore: ASF = LSF^2. For the passport photos: LSF = 3, so ASF = 3^2 = 9. Let us verify with real numbers. Area of standard passport print = 35 × 45 = 1 575 mm^2. Area of noticeboard print = 105 × 135 = 14 175 mm^2. Ratio of areas = 14 175 ÷ 1 575 = 9. ✓ This confirms ASF = 9 = 3^2. Practically, this means the noticeboard print uses exactly 9 times as much photographic paper as the passport print — even though its sides are only 3 times as long. This surprises many people, but it makes sense once you understand that area grows in two dimensions simultaneously. A farmer in Kericho who doubles the length and width of a tea plantation does not double its area — she quadruples it (2^2 = 4).

Section 4 — Identifying and Using Similar Figures in Real Life	
Describe at least TWO real-life situations — one from the passport photo context and one from a different Kenyan context — where identifying similar figures is useful or necessary. For each situation, explain how you would use the properties of similarity and the linear scale factor to solve a practical problem. Show at least one full worked calculation.	Situation 1 — Passport and ID photos: Passport regulations require that the face occupies a specific fraction of the photo height. If the standard 35 mm × 45 mm print shows a face height of 27 mm, and the studio needs to produce a 70 mm × 90 mm enlargement (LSF = 2), the face height in the enlarged photo must be 27 × 2 = 54 mm. A quality-check officer can verify similarity by confirming that 54 ÷ 27 = 90 ÷ 45 = 70 ÷ 35 = 2. If any of these ratios differ, the photo is distorted and the passport application will be rejected. Situation 2 — Map reading along the Nairobi–Mombasa highway: A geography student is using a road map with a scale of 1 : 1 000 000 (meaning 1 cm on the map represents 1 000 000 cm = 10 km in real life). The student measures the map distance between Nairobi and Voi as 32.6 cm. Using the linear scale factor: Real distance = 32.6 × 1 000 000 cm = 32 600 000 cm = 326 km. The student can also work in reverse: if the real distance from Voi to Mombasa is 160 km = 16 000 000 cm, then the map distance = 16 000 000 ÷ 1 000 000 = 16 cm. Both calculations rely directly on the property of similar figures: map and landscape are similar, with LSF = 1 000 000, so every corresponding length is in the ratio 1 : 1 000 000.

Section 5 — Reflect on Your Learning Journey	
Look back at your prediction	At the start of this unit, I predicted that resizing a photo preserved its shape simply because 'the computer copies it correctly' — I

from Lesson 1 and compare it to your explanation now. What did you think at the start about why resizing preserves shape? What key idea changed or deepened your thinking the most? What questions do you still have — about similarity, enlargement, or how these ideas extend to 3D solids?	had no mathematical explanation for why or how this happened. I could see that the noticeboard photo looked the same as the passport photo, but I could not explain what rule the software was following or why changing only one dimension breaks the image. The key idea that most changed my thinking was the Linear Scale Factor — the realisation that a single number can control an entire enlargement by being applied equally to every length. Before this unit, I thought 'making something bigger' was a vague process; now I understand it as a precise, testable mathematical operation. The relationship between the linear scale factor and the area scale factor ($ASF = LSF^2$) also genuinely surprised me. I had assumed that tripling the size of a photo meant using three times as much paper — but it actually uses nine times as much. This has real-world consequences I had never considered, like the cost of printing large posters or the amount of fabric needed to make a larger school uniform. Questions I still have: How does similarity extend to three-dimensional objects? If a sculptor in Kisumu creates a small clay model of a monument and then scales it up, does the volume scale by LSF^3? I also wonder: are there situations where only angles are equal but sides are NOT in ratio — and what is that type of figure called? I look forward to exploring volume scale factors and 3D similarity in future lessons.

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Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Answering the Driving Question (Conceptual Understanding)	Gives a precise, complete answer that correctly names similarity and the linear scale factor; clearly explains that uniform scaling of every length preserves all angles and all ratios; uses the passport photo numbers accurately and fluently.	Correctly identifies similarity and the linear scale factor; explains that all lengths are scaled equally; uses the passport photo context with mostly accurate numbers.	Mentions similarity or scale factor but explanation is vague or incomplete; may state a correct definition without connecting it to the passport photo phenomenon.
Conditions for Similarity and the Distortion Argument	States both conditions for similarity precisely (equal angles AND equal ratios of corresponding sides); gives a compelling, mathematically correct explanation of why changing only one dimension causes distortion; uses a second Kenyan example effectively.	States both conditions correctly; explains why distortion occurs when only one dimension changes; second example is relevant but may lack full mathematical detail.	States one condition for similarity or confuses the two conditions; explanation of distortion is present but lacks mathematical reasoning.
Linear Scale Factor and Area Scale Factor (Mathematical Reasoning)	Correctly calculates $LSF = 3$ and $ASF = 9$ from the passport photo measurements; derives or clearly shows why $ASF = LSF^2$ using algebraic or numerical reasoning; verifies with real area calculations (1 575 mm^2 and 14 175 mm^2); draws a meaningful practical conclusion.	Correctly calculates LSF and ASF; states the relationship $ASF = LSF^2$ and verifies it with the passport photo numbers; practical conclusion is present.	Calculates LSF correctly but makes errors with ASF; states the relationship $ASF = LSF^2$ without verification or with calculation errors.
Application to Real-Life Kenyan Contexts	Presents two clearly distinct, Kenyan-relevant situations; in each case correctly identifies the LSF and uses it to solve a practical problem; at least one full worked calculation is shown with correct units and logical steps.	Presents two relevant situations; uses LSF correctly in at least one full worked calculation; Kenyan context is clear and appropriate.	Presents one situation or two situations without a complete worked calculation; Kenyan context may be weak or generic.
Reflection and Scientific Thinking	Clearly contrasts initial prediction with	Contrasts initial and final understanding;	Reflection is present but vague; does not

	current understanding; identifies a specific key idea that changed thinking and explains WHY it was significant; poses at least one mathematically meaningful new question (e.g., volume scale factor, 3D similarity) that shows forward thinking.	identifies a key idea that developed across the unit; poses a new question that is relevant to the topic.	clearly identify what changed in thinking or new question is trivial or unrelated to the mathematics.